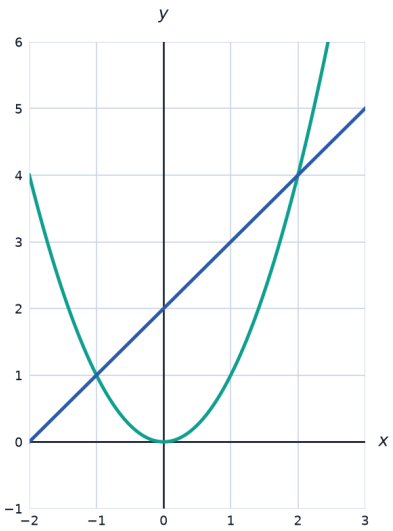
Area Between Curves

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

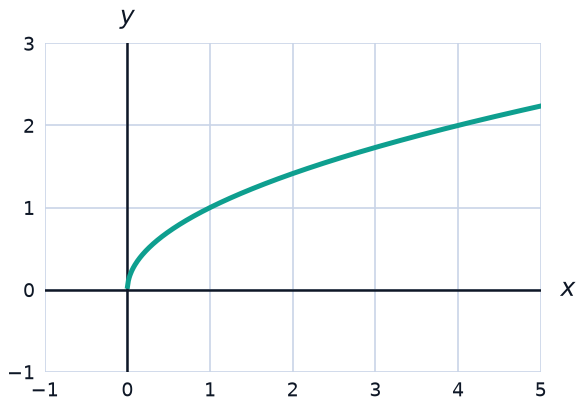
1 [Foundational] • Target: Area between two curves

Find the area of the region bounded by $y = x^2$ and $y = x + 2$.



2 [Foundational] • Target: Area between two curves

The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.



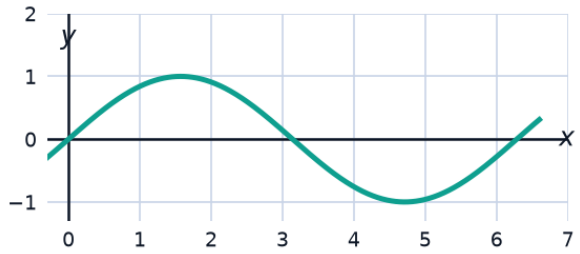
3 [Foundational] • Target: Setting up with respect to y, and total area

Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

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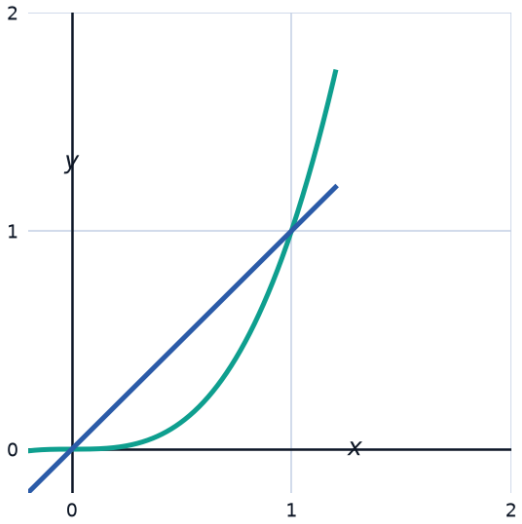
4 [Foundational] • Target: Setting up with respect to y, and total area

Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).



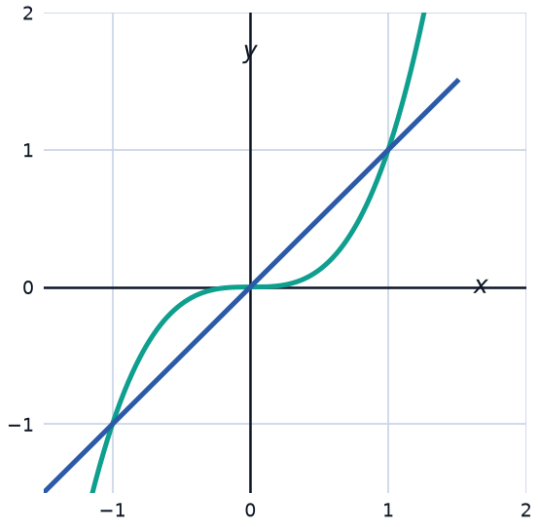
5 [Foundational] • Target: More area problems

Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).



6 [Foundational] • Target: Areas requiring case splits

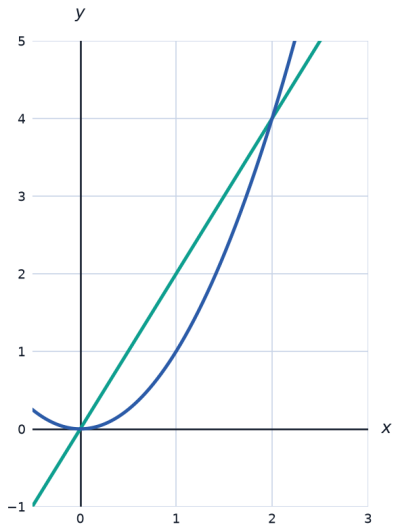
Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).



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7 [Foundational] • Target: Areas requiring case splits

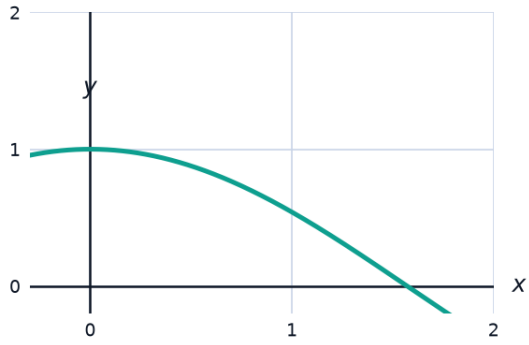
Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Areas involving trigonometric and exponential curves

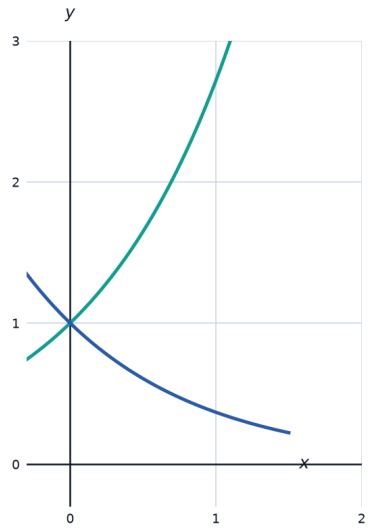
Find the area between $y = \cos x$ and $y = 0$ on $[0, \frac{\pi}{2}]$.



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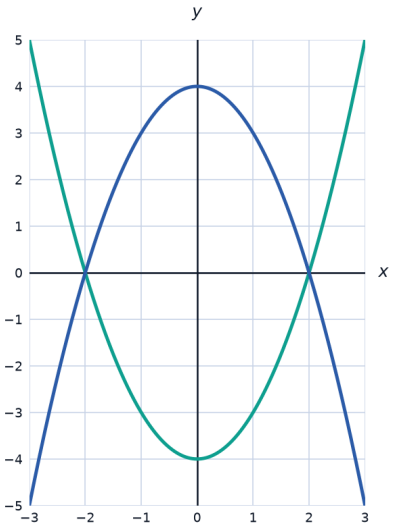
9 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).



10 **[Intermediate]** • **Target: Visualizing a region bounded by two curves**

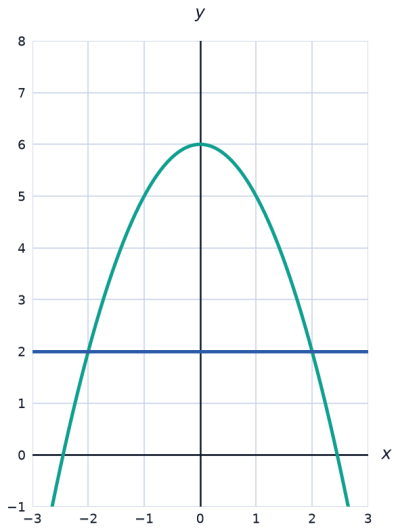
The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2)\,dx$.



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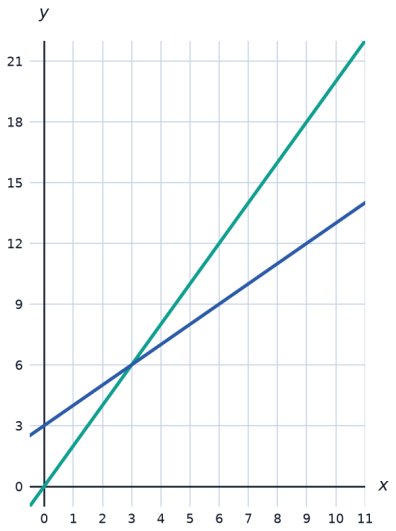
11 **[Intermediate]** • **Target: Area between a curve and a horizontal line**

Find the area between $y = 6 - x^2$ and $y = 2$.



12 **[Intermediate]** • **Target: Area between curves in applied contexts**

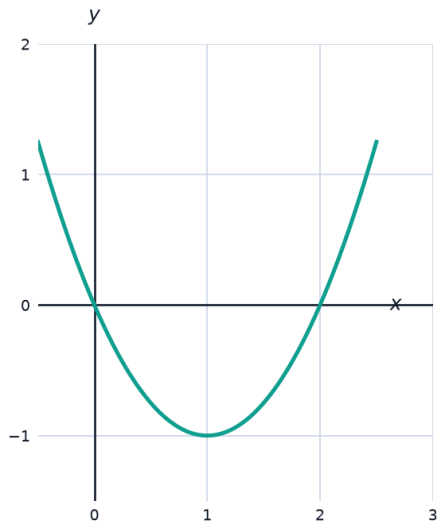
Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.



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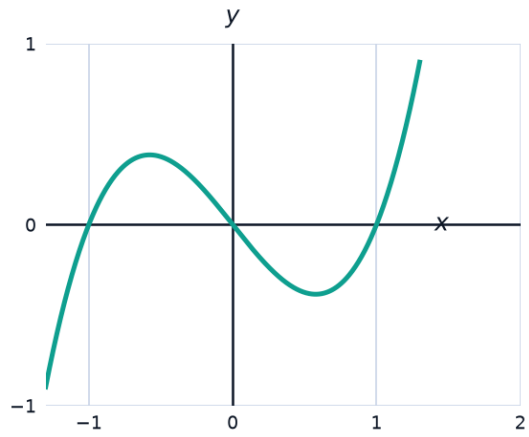
13 [Intermediate] • Target: Area between curves in applied contexts

Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).



14 [Intermediate] • Target: Area between curves in applied contexts

Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 3.01 The Derivative from First Principles

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

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16 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

17 [Challenging] • Target: Area between two curves • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

18 [Challenging] • Target: Setting up with respect to y, and total area • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

19 [Challenging] • Target: More area problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

20 [Challenging] • Target: Areas requiring case splits • Cross-Lesson: 6.01 Related Rates

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Area between two curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

22 [Foundational] • Target: Setting up with respect to y, and total area

Parametric area: region bounded by $x(t), y(t)$ from $t = a$ to $t = b$: $A = \int y\,dx =$

a. $\int_a^b y(t)x'(t)dt$

b. $\int_a^b x(t)y'(t)dt$

c. $\int_a^b y(t)dt$

d. $\int_a^b x(t)dt$

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23 [Foundational] • Target: More area problems

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e} - 2e - 1}{2e}$

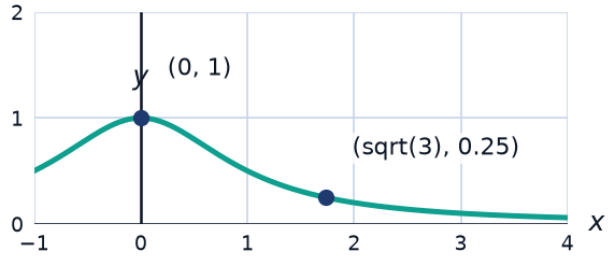
b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

24 [Foundational] • Target: Areas requiring case splits

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?



a. $\sqrt{2}$

b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$

d. 2

25 [Foundational] • Target: Areas involving trigonometric and exponential curves

Polar form of $y = c$ (horizontal line):

a. $r = c\csc\theta$

b. $r\cos\theta = c$

c. $r = c$

d. $r\sin\theta = c$

26 [Foundational] • Target: Visualizing a region bounded by two curves

Centroid $\bar{y}$ of the region bounded by $y = x^2$ and $y = 4$:

a. $\bar{y} = 5/2$

b. $\bar{y} = 8/5$

c. $\bar{y} = 2$

d. $\bar{y} = 12/5$

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27 [Foundational] • Target: Area between a curve and a horizontal line

Volume of region bounded by $z = 1 - x^2 - y^2$ ($z \geq 0$):

a. $\pi/4$

b. $\pi/2$

c. π

d. $3\pi/4$

28 [Foundational] • Target: Area between curves in applied contexts

Area of polar region swept by $r = \theta$ from 0 to 2π:

a. $\pi^3/3$

b. $4\pi^3/3$

c. $2\pi^3/3$

d. π^2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Area between two curves

Evaluate $\iint_R x\,dA$ where R is the region between $y = x^2$ and $y = x$.

a. $\frac{1}{6}$

b. $\frac{1}{24}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

30 [Intermediate] • Target: Setting up with respect to y, and total area

The area of the region enclosed by $x = t - \sin t$, $y = 1 - \cos t$ (one arch of cycloid) using $A = \int y\,dx$:

a. πr^2

b. $3\pi r^2$ (with $r = 1$)

c. π

d. 2π

31 [Intermediate] • Target: More area problems

Line of intersection of two non-parallel planes: direction vector $=$

a. sum of normals

b. cross product of normals

c. difference of normals

d. dot product of normals

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32 **[Intermediate]** • **Target: Areas requiring case splits**

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

33 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

34 **[Intermediate]** • **Target: Visualizing a region bounded by two curves**

What is the area of the region bounded by the polar curve $r = \sqrt{5\sin\theta}$ for $0 \le \theta \le \pi$?

a. 5

b. 10

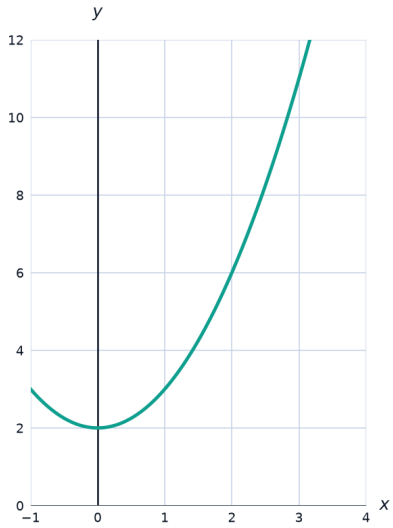
c. $\frac{5}{2}$

d. 5π

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35 **[Intermediate]** • **Target: Area between a curve and a horizontal line**

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?



a. $\int_0^3 (x^2 + 2)\,dx$

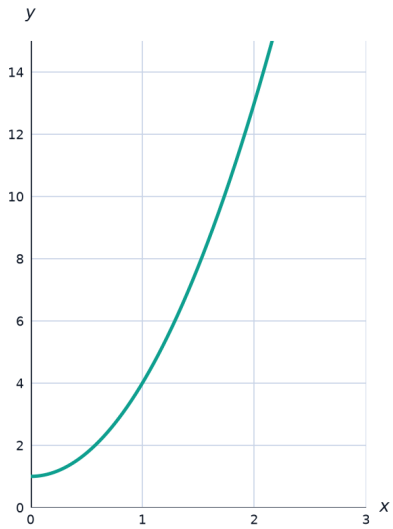
b. $\int_0^3 (x^2 - 2)\,dx$

c. $\int_0^3 x^2\,dx$

d. $\int_0^3 2\,dx$

36 **[Intermediate]** • **Target: Area between curves in applied contexts**

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?



a. 10

b. 12

c. 14

d. 16

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37 [Intermediate] • Target: Area between two curves

Green's first identity states (for smooth u, v on bounded Ω):

a. $\int_\Omega ablau \cdot ablav\, dV = \oint_{\partial\Omega} u \frac{\partial v}{\partial n} dS - \int_\Omega u\, \Delta v\, dV$

b. $\int_\Omega u\, \Delta v\, dV = \oint_{\partial\Omega} u \frac{\partial v}{\partial n} dS - \int_\Omega ablau \cdot ablav\, dV$

c. $\int_\Omega u\, \Delta v = \int_\Omega v\, \Delta u$

d. $\oint u\, \Delta v\, dS = \int_\Omega ablau \cdot ablav\, dV$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Setting up with respect to y, and total area • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

The total differential: $df = f_x\, dx + f_y\, dy + f_z\, dz$. For $f = xyz$, $df =$

a. $xyz(dx + dy + dz)$

b. $yz\, dx + xz\, dy + xy\, dz$

c. $y\, dx + z\, dy$

d. $x\, dx + y\, dy + z\, dz$

39 [Challenging] • Target: More area problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Level curve: $f(x, y) = c$. The gradient $ablaf$ is $1cm$ to the level curve.

a. tangent

b. perpendicular (normal)

c. parallel

d. zero

40 [Challenging] • Target: Areas requiring case splits • Cross-Lesson: 6.01 Related Rates

Exponential growth model: if $\frac{dP}{dt} = kP$, then $P(t) =$

a. $P_0 + kt$

b. $P_0 kt$

c. ke^t

d. $P_0 e^{kt}$

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41 [Challenging] • Target: Areas involving trigonometric and exponential curves • Cross-Lesson: 6.02 Optimization and Curve Sketching

Horizontal tangent to $y = x^3 - 3x$ at:

a. $x = 0$

b. $x = \pm 1$

c. $x = \pm\sqrt{3}$

d. $x = 3$

42 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 7.01 Integration by Substitution

Cauchy MVT applied to f, g: gives $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$. It is used to derive:

a. Rolle's theorem

b. L'Hôpital's rule

c. IVT

d. EVT

43 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 7.02 Integration by Parts

Complex exponential: $e^{i\theta} =$

a. $\sin\theta + i\cos\theta$

b. $\cos\theta - i\sin\theta$

c. $\cosh\theta + i\sinh\theta$

d. $\cos\theta + i\sin\theta$

44 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 8.02 Volumes of Revolution

Partial with respect to z: $f(x, y, z) = xyz^2$, $f_z =$

a. $2xyz$

b. xyz

c. yz^2

d. xy

45 [Challenging] • Target: Area between two curves • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Torsion $\tau = -d\mathbf{B}/ds \cdot \mathbf{N}$. For a plane curve, $\tau =$

a. 1

b. 0

c. κ

d. -1

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